

Leif Hellgren

Newport Beach, CA (relocating to New York City, September 2026)

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SUMMARY

Physics-trained engineer who owned a flight-critical subsystem end to end at Blue Origin, from first-principles modeling through hardware delivery and test. I build my own software rather than waiting on someone else's tools. Since then I've spent 18 months building Python and LLM-driven software independently, and shipped production features into a live Next.js codebase in a one-month contract. Comfortable across systems engineering, data pipelines, and full-stack work. Relocating to New York City in September 2026.

EXPERIENCE

Independent Developer, Python & LLM Systems | Self-Employed

November 2024 - April 2026

- Built a full software stack solo for stock market analysis: a Streamlit analytics application over Pandas tracking performance and risk metrics in real time, a scanner filtering thousands of tickers against defined criteria, and a staged alerting pipeline with webhook-based delivery.
- Used LLMs throughout the build process for design, debugging, and iteration, and wrote a 26-section technical specification with an 8-phase validation plan before shipping anything to production use.
- Built a Python job-board monitor integrating 3 ATS APIs (Ashby, Greenhouse, Lever) across 27 companies, with tiered relevance classification, deduplication against a persistent state store, and scheduled automation.
- Reverse-engineered an undocumented HTML export format to build an automated resume-tailoring pipeline with headless Chrome PDF rendering, diagnosing and fixing an escaping bug that would have silently corrupted output files.

Growth & Engineering (contract) | Launchpoint

June - July 2026

Case study: building the outbound motion and CRM from scratch (read [here](#))

- Shipped a new feature into a live production codebase (Next.js/TypeScript/Drizzle ORM) with no ramp time: media performance metrics (spend, impressions, CTR, CPC, ROAS) surfaced in a dashboard modal, plus a scoring model to predict ad quality before spend.

Responsible Engineer, Flight-Critical Cryogenic Fluid Management System (Lunar Lander) | Blue Origin

June 2022 - November 2024 | El Segundo, CA

- Owned a flight-critical cryogenic pressure management subsystem on Blue Origin's first lunar lander, from concept through hardware integration and test readiness, and delivered it through PDR, CDR, and TRR.
- Built first-principles fluid and thermal models in Python and MATLAB to size components and run trade studies, so design decisions didn't wait on formal analysis cycles.
- Led the subsystem FMECA and an independent hydrogen accumulation and ignition risk analysis, and the mitigations went into the vehicle design.
- Coordinated seven teams (thermal, GNC, structures, manufacturing, integration, safety, test) and managed vendors end to end, from specs through qualification and acceptance testing.

LEADERSHIP

Founder, Aerospace Research Cooperative. Co-Founder & Lead Propulsion Feed System Engineer, UCSB Rocket Propulsion Laboratory (2017-2021). Founded UCSB's first aerospace organization and co-founded what became the university's largest engineering organization.

EDUCATION

Bachelor's in Physics | University of California, Santa Barbara

2017-2021

SKILLS

Software & Data: Python (Pandas, NumPy, Streamlit), TypeScript/Next.js, SQL, Drizzle ORM, Git, MATLAB, Pine Script

Engineering: systems engineering, mechanical design (CAD, P&IDs, BOMs), fluid and thermal analysis, first-principles modeling, FMECA and risk analysis